# CLASSIFICATION & EQUILIBRIUM

- ORDER: NUMBER OF STATES EQUATIONS
- SISO, SIMO, MISO, MIMO: NUMBER OF INPUT AND OUTPUT
- DYNAMIC/STATIC: (STATE)
  - DYNAMIC: STATE PRESENT IN OUTPUT EQUATION
  - STATIC: STATE ABSENT IN OUTPUT EQUATION
- TIME VARIANCE/INVARIANCE: (TIME)
  - TIME VARIANT: TIME PRESENT IN OUTPUT EQUATION
  - TIME INVARIANT: TIME ABSENT IN OUTPUT EQUATION
- LINEAR/NOT-LINEAR:
  - LINEAR: FUNCTION $f$ AN $g$ ARE LINEAR (FIRST DEGREE)
  - NOT LINEAR: $f$ AND $g$ ARE NOT LINEAR
- PROPER/STRICTLY PROPER: (INPUT)
  - PROPER: INPUT PRESENT IN OUTPUT EQUATION
  - STRICTLY PROPER: INPUT ABSENT IN OUTPUT EQUATION

- LINEAR SYSTEM:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$
$$y(t) = C(t)x(t) + D(t)u(t)$$

- LINEAR TIME INVARIANT (LTI) SYSTEM:

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$y(t) = Cx(t) + Du(t)$$

- PROPER / STRICTLY PROPER
  - INPUT PRESENT IN OUTPUT EQ.
  - INPUT ABSENT IN OUTPUT EQ.
  - ORDER N = ORDER D
  - ORDER N $\neq$ ORDER D

## SYSTEMS

$$\dot{x}_1(t) = x_1^2(t)(x_2(t) - 2) + u(t)$$

$$\dot{x}_2(t) = (x_1(t) - \alpha)(x_2(t) - 2) - x_2(t)$$

$$y(t) = x_1(t)\, x_2(t)$$

- **EVALUATE EQUILIBRIUM STATE CONNECTED TO THE INPUT**
  - "CONNECTED TO THE INPUT" MEANS $\bar{w}=0$
  - SO I HAVE EQ $((\bar{x}_1, \bar{x}_2), 0)$ ← I NEED TO FIND $\bar{x}_1$ AND $\bar{x}_2$

    IF THERE IS NO EQU. POINT WE WRITE "NO FEASIBLE EQU. POINT"
  - IN EQUILIBRIUM ALSO
    - $\dot{x}_1(t)=0$
    - $\dot{x}_2(t)=0$
  - REWRITE THE SYSTEM WITH THOSE NEW INFORMATIONS

    $\begin{cases} \bar{w}=0 \leftarrow \text{REMOVE INPUT FROM THE STATES} \\ \dot{x}_1(t)=0 \\ \dot{x}_2(t)=0 \end{cases}$
  - FIND THE VALUES FOR $x_1(t)$ AND $x_2(t)$ THAT SATISFY THE SYSTEM EQ.
  - THOSE VALUES ARE $\bar{x}_1$ AND $\bar{x}_2$
  - REWRITE EQ $((\bar{x}_1; \bar{x}_2), w)$ WITH THE CORRECT VALUES
- **DETERMINE LINEARIZE SYSTEM ON THE EQUILIBRIUM POINTS**
  - $\delta w = w(t) - \bar{w}$
  - $\delta \dot{x}(t) = x(t) - \bar{x}$ → $\delta \dot{x}_1(t) = x_1(t) - \bar{x}_1$ ; $\delta \dot{x}_2(t) = x_2(t) - \bar{x}_2$
  - $\delta y(t) = y(t) - \bar{y}$

    INSTEAD OF $x_1(t)$, $x_2(t)$ AND $w(t)$ USE $\bar{x}_1$, $\bar{x}_2$ AND $\bar{w}$
  - COMPUTE $\bar{y}$ BY USING THE FORMULA FOR $y(t)$ PROVIDED FROM THE EXERCISE
  - SUBSTITUTE IN THE EQUATIONS WHAT WE KNOW $\longrightarrow \bar{x}_1, \bar{x}_2, \bar{w}, \bar{y}$
  - REWRITE $\delta x_1(t)$, $\delta x_2(t)$, $\delta w(t)$ AND $\delta y(t)$ $\longrightarrow$ DO NOT CHANGE $x_1(t), x_2(t), y(t), w(t)$
- **STUDY THE STABILITY OF THE EQUILIBRIUM POINTS** ← ONLY IF THE EQU. POINT EXISTS
  - REMEMBER: $\begin{cases} \dot{x}(t) = Ax(t) + Bw(t) \\ y(t) = Cx(t) + Dw(t) \end{cases}$
  - IF WE HAVE TWO STATE EQUATIONS $\dot{x}_1(t) = f_1(\underline{x}, \underline{w})$ AND $\dot{x}_2(t) = f_2(\underline{x}, \underline{w})$
  - THE OUTPUT EQUATION WILL BE $y(t) = g(\underline{x}, \underline{w})$
  - I NEED TO FIND: A, B, C, D

$$A = \begin{bmatrix} \frac{df_1(\underline{x},\underline{w})}{dx_1} & \frac{df_1(\underline{x},\underline{w})}{dx_2} \\ \frac{df_2(\underline{x},\underline{w})}{dx_2} & \frac{df_2(\underline{x},\underline{w})}{dx_2} \end{bmatrix} \; ; \; B = \begin{bmatrix} \frac{df_1(\underline{x},\underline{w})}{dw} \\ \frac{df_2(\underline{x},\underline{w})}{dw} \end{bmatrix}$$

(# VARIABLES ; # STATE EQUATIONS) — (# INPUT X ; # STATE EQUATIONS), # INPUT → USUALLY 1

IF IN THE MATRICES WE HAVE STILL $x_1(t)$, $x_2(t)$, $w(t)$ WE SUBSTITUTE THEM WITH $\bar{x}_1, \bar{x}_2, \bar{w}$

$$C = \begin{bmatrix} \frac{dg(\underline{x},\underline{w})}{dx_1} ; \frac{dg(\underline{x},\underline{w})}{dx_2} \end{bmatrix} \; ; \; D = \begin{bmatrix} \frac{dg(\underline{x},\underline{w})}{dw} \end{bmatrix}$$

(# VARIABLES ; # OUTPUT FUNCTIONS), # OUTPUT FUNCTIONS → USUALLY 1 — (# INPUT ; # OUTPUT FUNCTIONS), # INPUT → USUALLY 1, # OUTPUT FUNCTIONS → USUALLY 1

- WRITE: $\begin{cases} \delta \dot{x}_1(t) = A\,\delta \underline{x}(t) + B\,\delta \underline{w}(t) \\ \delta \dot{x}_2(t) = A\,\delta \underline{x}(t) + B\,\delta \underline{w}(t) \\ \delta y(t) = C\,\delta \underline{x}(t) + D\,\delta \underline{w}(t) \end{cases}$

  BASICALLY JUST SOLVE THE MATRICES OF THE FORMULAS. REMEMBER $\delta$ IN FRONT OF $x_1(t)$ AND $x_2(t)$ WHEN SOLVING THE MATRICES!

- COMPUTE $\det(\lambda I - A)$ TO FIND THE EIGENVALUES AND SET $\det(\lambda I - A) = 0$
  - REMEMBER: $\det\begin{pmatrix} A & B \\ C & D \end{pmatrix} = AD - BC$
- CASES:
  - IF $\lambda_1 < 0$ AND $\lambda_2 < 0$ $\longrightarrow$ SYSTEM ASYMPTOTICALLY STABLE
  - IF $\lambda_1 = 0$ AND $\lambda_2 < 0$ (OR VICE-VERSA) $\longrightarrow$ SYSTEM SIMPLY STABLE
  - IF $\lambda_1 > 0$ OR $\lambda_2 > 0$ $\longrightarrow$ SYSTEM UNSTABLE

## SYSTEMS LTI

ORDINE MATRICI: DA SINISTRA A DESTRA

$$\begin{cases} \dot{x}_1(t) = x_1(t) - 3x_2(t) + u_1(t) \\ \dot{x}_2(t) = x_1(t) + x_2(t) + u_2(t) \\ \\ y(t) = x_1(t) + u_1(t) \end{cases}$$

- **DEFINE TRANSFER FUNCTION**
  - THE TRANSFER FUNCTION IS: $G(s) = C(sI - A)^{-1}B + D$
  - I NEED TO FIND: A, B, C, D
  - REMEMBER: $\begin{cases} \dot{x}(t) = Ax(t) + Bw(t) \\ y(t) = Cx(t) + Dw(t) \end{cases}$

$$A = \begin{bmatrix} 1 & -3 \\ 1 & 1 \end{bmatrix} ; \; B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} ; \; C = [1, 0] ; \; D = [1, 0]$$

REMEMBER: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

- COMPUTE $G(s)$

$$(sI - A) = \begin{bmatrix} s-1 & 3 \\ -1 & s-1 \end{bmatrix} \longrightarrow (sI - A)^{-1} = \frac{1}{(s-1)(s-1)+3} \begin{bmatrix} s-1 & -3 \\ 1 & s-1 \end{bmatrix}$$

$$C(sI - A)^{-1} = \frac{1}{s^2 - 2s + 4} [1, 0] \begin{bmatrix} s-1 & -3 \\ 1 & s-1 \end{bmatrix} = \frac{1}{s^2-2s+4} [s-1+0; -3+0] = \frac{1}{s^2-2s+4} [s-1; -3]$$

$$C(sI - A)^{-1}B = \frac{1}{s^2-2s+4} [s-1; -3] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{s^2-2s+4} [s-1; -3]$$

$$C(sI - A)^{-1}B + D = \frac{1}{s^2-2s+4} [s-1; -3] + [1; 0] = \left[ \frac{s-1}{s^2-2s+4} + 1 ; \frac{-3}{s^2-2s+4} + 0 \right]$$

- **IS THE TRANSFER FUNCTION A SOLID APPROXIMATION OF THE SYSTEM? MOTIVATE IT**
  - YES, AS FOR BOTH TRANSFER FUNCTION, BECAUSE NO CANCELLATIONS WERE INVOLVED
  - ALSO IF THE SEMPLIFICATION INVOLVES A STABLE POINT $(s<0)$ IS STABLE

## RESPONSE & BODE DIAGRAM

| $1^{st}$ Order Strictly Proper | | | | $1^{st}$ Order Proper | $2^{nd}$ Order: complex pole | |
|---|---|---|---|---|---|---|
| $T_s$ | $T_r$ | $T_{a5}$ | $T_{a1}$ | $T_{a\varepsilon}$ | $S\%$ | $T_{a\varepsilon}$ |
| $\approx 2.2T$ | $\approx 0.7T$ | $\approx 3T$ | $\approx 4.7T$ | $T\ \ln\frac{\lvert 1-\alpha\rvert}{0.01\varepsilon}$ | $100e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$ | $-\frac{\ln 0.01\varepsilon}{\xi\omega_n}$ |

### PREPROCESSING

**a) CHECK STABILITY**

P = POLES → NUMBERS THAT MAKES DENOMINATOR = 0

Z = NUMBER THAT MAKE NUM. = 0

IF POLES > 2 SPECIFY IF REAL OR NOT AND DISTINCT OR NOT

- IF $\forall\ P<0$ → ASYNTOTICALLY STABLE
- IF $\exists\ P=0$ → STABLE
- IF $\exists\ P>0$ → UNSTABLE

$\nu$ = ZEROS IN P - ZEROS IN Z

**b) ORDER OF SYSTEM**

\# OF POLES → ORDER
(DENOMINATOR)

- IF # P = 1 → FIRST ORDER
- ECC...

**c) PROPER/STRICTLY PROPER** ← ONLY FOR 1° ORDER

- ORDER N = ORDER D → PROPER
- ORDER N ≠ ORDER D → STRICTLY PROPER

ORDER: NUMBER OF ZEROS

**d) CONCLUSION**

WRITE
- STABILITY
- ORDER ← ONLY FOR 1° ORDER
- PROPER/STRICTLY PROPER

### • FIND EQUILIBRIUM VALUES

- IF STABLE → USE FINAL VALUE THEOREM (FVT)

$$\nu = \lim_{s\to 0} s^{\nu} G(s)U(s)$$

UNIT STEP: 1 IN GENERAL

$$U(s) = \begin{cases} \text{STEP}(t) = \frac{U}{s} \\ \text{RAMP}(t) = \frac{U}{s^2} \\ e^{\omega t}\,\text{STEP}(t) = \frac{U}{s-\omega} \\ \sin(\omega t)\,\text{STEP}(t) = \frac{U\omega}{s^2+\omega^2} \\ e^{\omega t}\sin(\omega t)\,\text{STEP}(t) = \frac{U\omega}{(s-\omega)^2+\omega^2} \end{cases}$$

← USUALLY IS THIS, CHECK THE VALUE OF ω

### • SETTLING TIME AT 5%

CHECK THE TYPE OF SYSTEM

**• IF 1° ORDER, STRICTLY PROPER**

$T_{a5} \approx 3T \approx 3\left|\frac{1}{p}\right| \approx \ldots s$

SECONDS!

POLES

**• IF 1° ORDER, PROPER**

$T_{a\varepsilon} = T\ln\frac{|1-\alpha|}{0.01\varepsilon}$ $\quad T = \left|\frac{1}{p}\right|$ $\quad \tau = \frac{1}{z}$

$\alpha = \frac{\tau}{T} = \frac{1/z}{|1/p|} = \frac{|p|}{z}$

$\varepsilon = 5$ ← % GIVEN FROM TEXT

**• IF $2^{ND}$ ORDER, REAL**

IF ALL THE POLES ARE REAL

USE DOMINANT POLE APPROXIMATION. FIND THE DOMINANT POLE (AKA THE ONE CLOSEST TO ZERO → AKA $T = \left|\frac{1}{p}\right|$ BIGGEST)

- THEN APPROXIMATES TO FIRST ORDER STRICTLY PROPER WITH ONLY 1 POLE (THE BIGGEST)

**• IF $2^{ND}$ ORDER COMPLEX POLE**

$T_{a\varepsilon} = \frac{-\ln(0.01)\varepsilon}{\xi\omega_n}$

TO FIND ξ AND $\omega_n$ REMEMBER: $G(s) = \frac{\nu\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$

↳ $\xi\omega_n = \frac{\text{COEF}(s)}{2}$

### • OVERSHOOTING %

CHECK THE TYPE OF SYSTEM

**• IF 1° ORDER, STRICTLY PROPER**

NO OVERSHOOT

**• IF 1° ORDER, PROPER**

$\alpha = \frac{|p|}{z}$

- IF $\alpha < 1$ NO OVERSHOOT, BUT RAISE
- IF $\alpha > 1$ OVERSHOOT

**• IF $2^{ND}$ ORDER, REAL**

IF ALL THE POLES ARE REAL

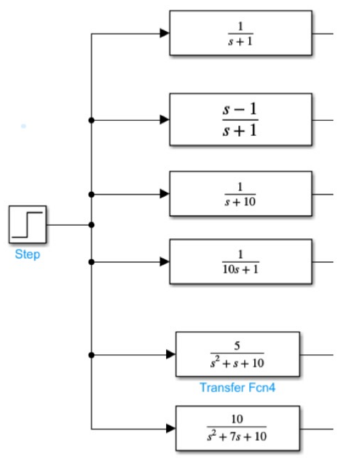
SINCE I APPROXIMATED TO A $1^{ST}$ ORDER STRICTLY PROPER THERE IS NO OVERSHOOT

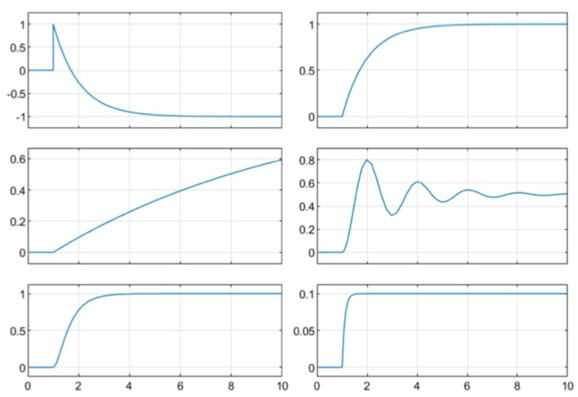
**• IF $2^{ND}$ ORDER COMPLEX POLE**

$$S\% = 100e^{\frac{-\xi\pi}{\sqrt{1-\xi^2}}}$$

# CONNECT SYSTEM AND STEP RESPONSE

Connect the systems with the correct step response



TO SOLVE THOSE TYPE OF EXERCISES YOU NEED TO COMPUTE THE EQUILIBRIUM POINT OF EACH STEP RESPOSE.

## STEPS

### 1) CLASSIFY THE SYSTEM

- POLES, Z
- STABLE/UNSTABLE ?
- PROPER/STRICTLY PROPER
- $1^{ST}, 2^{ND}, \dots$ ORDER

### 2) EQUILIBRIUM POINT

$$N = \lim_{s \to 0} \cancel{s} \cdot G(s) \boxed{\frac{1}{\cancel{s}}}$$

SINCE STEP RESPONSE

N IS THE LIMIT OF THE SYSTEM (WHERE IT IS HEADED TO)

### 3) TIME CONSTANT (IF EQUILIBRIUM NOT ENOUGH)

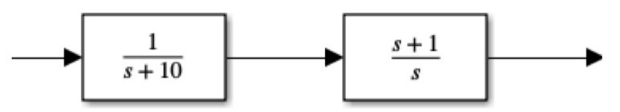
- COMPUTE $T = \left|\frac{1}{p}\right|$
- A SMALLER $T$ MEANS THAT THE SYSTEM REACH THE EQUILIBRIUM FASTER

## REMEMBER:

- IF A SYSTEM IS STRICTLY PROPER WE WILL HAVE AN ISTANTANEOUS RISE
- A $2^{ND}$ ORDER SYSTEM IS STABLE IF BOTH $s_1, s_2$ HAVE A REAL PART LESS THAN 0
- IF PROPER (DEG N. = DEG D.) WE HAVE GRAPH
- $2^{ND}$ ORER COMPLEX WE HAVE OSCILLATIONS ONLY IF $\xi < 0.07$

# EVALUATE THE SIGNAL

Given the systems



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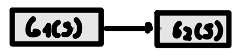
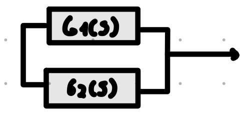


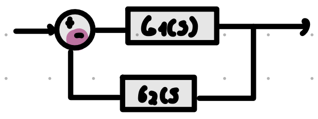
$G_{TOT}(S) = G_1(S)\cdot G_2(S)$

PARALLEL:



$G_{TOT}(S) = G_1(S) + G_2(S)$

FEEDBACK:



$G_{TOT}(S) = \dfrac{G_1(S)}{1 \pm G_1(S)\cdot G_2(S)}$

+ FOR NEGATIVE FEEDBACK
- FOR POSITIVE FEEDBACK

## EVALUATE THE RESPONSE OF THE SIGNAL $10\sin(10t)$

0) SINCE THE SYSTEM IS IN SERIES THAN: $G_{TOT}(S) = G_1(S)\cdot G_2(S)$

$$G_{TOT}(S) = \frac{1}{S+10}\cdot\frac{S+1}{S} = \frac{S+1}{S^2+10S}$$

1) FIND W AND COMPUTE $G(JW)$

$W =$ "CONSTANT INSIDE SIN/COS" $= 10$

$$G(JW) = G(J10) = \frac{10J+1}{100J^2+100J} = \frac{10J+1}{-100+100J}$$

2) $|G(JW)|$ COMPUTE MODULE OF $G(JW)$

$$|G(JW)| = \left|\frac{10J+1}{-100+100J}\right| = \frac{|10J+1|}{|100J-100|} = \frac{\sqrt{10^2+1^2}}{\sqrt{100^2+100^2}} = \frac{\sqrt{101}}{\sqrt{20000}} = 0.071$$

3) FIND PHASE $\varphi$

$$\varphi = \measuredangle G(JW) = \measuredangle G(10J) = ARG\left(\frac{10J+1}{100J-100}\right) = ARG(10J+1) - ARG(100J-100)$$

$$= TAN^{-1}\left(\frac{10}{1}\right) - TAN^{-1}\left(\frac{100}{100}\right) \approx 90° - 135° = -45°$$

4) MERGE ALL

$$|G(JW)|\,10\sin(10t-\varphi) = 0.071\ 10\sin(10t-45)$$

REMEMBER:
- $J^2 = -1$
- $|a i + b| = \sqrt{a^2+b^2}$
- $\measuredangle G(JW) = ARG(G(JW))$
- $ARG(A/B) = ARG(A) - ARG(B)$
- $ARG(a i + b) = TAN^{-1}\left(\frac{a}{b}\right)$

IF THE SIGNAL IS IN TWO PARTS LIKE $\sin(100t) + 3\cos(t)$ WE CAN COMPUTE THEM SEPARATELY AND ADD THEM IN THE END

## DEFINE THE BODE DIAGRAM

EX: $P = \{1/10;\ 100+5j\}$ $Z = \{-3\}$ $\gamma = 1$ (IT SHOULD BE 0 IN THIS CASE) $(N>0)$

### MODULE GRAPH:

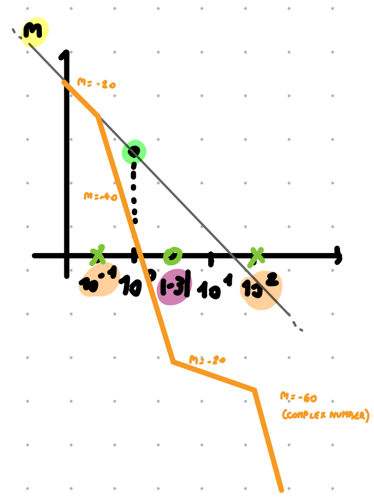
- COMPUTE $W \to 0$:

  $M = -20\gamma$

- AFTER FINDING M WE NEED TO FIND A POINT WHERE THE LINE PASSES, THAT IS:

  $(x,y) = (10^0;\ 20\log_{10}(N))$ $N = \lim_{S\to 0} S^\gamma G(S) U(S)$

- ON MY X-AXIS I ADD ALL THE P AND Z THAT I HAVE FOUND IN G(S). USE "x" FOR P AND "o" FOR Z
  - IF A NUMBER IS COMPLEX TAKE ONLY REAL PART
  - IF A NUMBER IS NEGATIVE TAKE THE ABSOLUTE VALUE
- NOW I NEED TO REDRAW MY LINE:
  - EVERY TIME IT MEETS A "x" (P) THE M DECREASE OF 20
  - EVERY TIME IT MEETS A "o" (Z) THE M INCREASE OF 20
  - IF THE "x" OR "o" IS A COMPLEX NUMBER THE M DECREASE/INCREASE OF 40




EX: $P = \{1/10;\ 100+5j\}$ $Z = \{-3\}$ $\gamma = 1$ (IT SHOULD BE 0 IN THIS CASE) $N>0$

### PHASE GRAPH:

- CHECK N:

  FROM EXAMPLE → $N>0$: $START(\varphi) = 0 - 90\gamma = -90\cdot 1 = -90$

  $N<0$: $START(\varphi) = \boxed{-}180 - 90\gamma$

- ON MY X-AXIS I ADD ALL THE P AND Z THAT I HAVE FOUND IN G(S). USE "x" FOR P AND "o" FOR Z
  - IF A NUMBER IS COMPLEX TAKE ONLY REAL PART
  - IF A NUMBER IS NEGATIVE TAKE THE ABSOLUTE VALUE
- NOW I DRAW MY LINE
  - EVERY TIME IT MEETS A "x" (P) POSITIVE THE LINE INCREASE OF 90
  - EVERY TIME IT MEETS A "x" (P) NEGATIVE THE LINE DECREASES OF 90
  - EVERY TIME IT MEETS A "o" (Z) POSITIVE THE LINE DECREASE OF 90
  - EVERY TIME IT MEETS A "o" (Z) NEGATIVE THE LINE INCREASES OF 90
  - IF THE "x" OR "o" IS A COMPLEX NUMBER THE LINE INCREASES/DECREASES OF 180